TG3D Street Library assembly instructions

The TG3D street library is a strong, robust box which enhances your community engagement.

The box is constructed of “C/D” grade 12mm plywood. The “C” side (the better quality side) is on the outside of the box to simplify painting. There will be marks and occasional top surface knots on the material which will not affect the functionality or durability of the box. Plywood is dimensionally stable and the box should slot together however a tiny bit of “encouragement” might be needed. The CNC cut is designed to be tight to assist in excluding water from the joints..

The box is machine cut using a “Computer Numeric Control” (CNC) router. This provides accurate joins and makes the boxes simple to assemble. It is screwed, not nailed, for strength, durability and water resistance. The screw holes are pre-drilled.

The kit is supplied with sufficient screws to assemble the box and fit the hardware supplied.

Please read this instruction completely before commencing assembly.

Tools required.

- #2 Phillips head screwdriver. A power screwdriver will make this job faster however make sure you turn the torque settings down. This also allows you to “set” the screws more accurately. Flush with the top surface is sufficient.
- #1 Phillips head screwdriver. This is used for the screws that hold the acrylic windows on.
- Sandpaper. 80 grit should be sufficient. Use this to sand the completed box prior to painting. A small orbital sander works well if you have one but hand sanding won’t take long. I give the external surfaces a basic sand before dispatch.

Parts

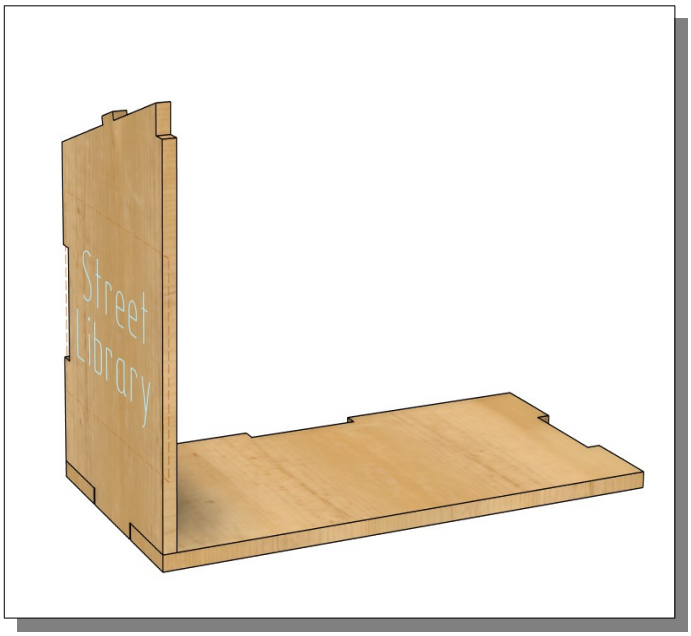
The box is supplied with the following parts

- Base
- 2 sides - there is a left and a right side. The front edge is the larger edge
- Back
- Top

- 2 doors – these are different to each other. The recessed sides are the inner sides and hold the acrylic. There is a hole near the inner edge to hold the handle. There is also 2 small “pops” hole on the inside lower edge. These marks the position of the male part of the catch. The catch goes on the base of the library.
- 2 handles. These are pre-drilled to centre them correctly and have a corresponding hole pre-drilled in the doors
- Top door jamb – note there is a small rebate on each end. This rebate faces outwards.
- 2 Acrylic window panes – there are 12 4G x 12mm screws to secure them.
- 2 305mm piano hinges – with 24 stainless screws in 2 packets.
- 1 spring type catch with a corresponding striker – it uses the screws supplied with them. Note: I supply 2 catches but only one is really needed.
- 35 6G x 20mm screws – there should be some spare! 2 of them are used to secure the handles.

Assembly

Start with the base. It has 3 notches in its profile and 2 small “tic” marks on the inside and front edge. These are used to accurately locate the ke



Take the left hand side (as shown) and push the tab in the side into the base. Make sure the side is pushed right down. Place a screw into the “tab” on the side and loosely tighten it. Turn the base and side onto their back edges and whilst aligning the side to the base, put the 2 screws in the base to secure the side.

The next step is to add the back to the library. Flip the base/side onto the “front edge” and insert the base. The bottom tab on the back engages with the notch on the base. The tab on the left hand side (looking from the front of the box) engages with the left hand side.



Put 2 screws into the tabs on the back making sure the back is firmly engaged into the base and side. Next step is 2 screws in the side that screw into the back and 4 screws in the base that connect to the back. At this point you can tighten all screws inserted so far. The 3 sides should also be getting quite rigid

Next step is the right hand side (RHS).

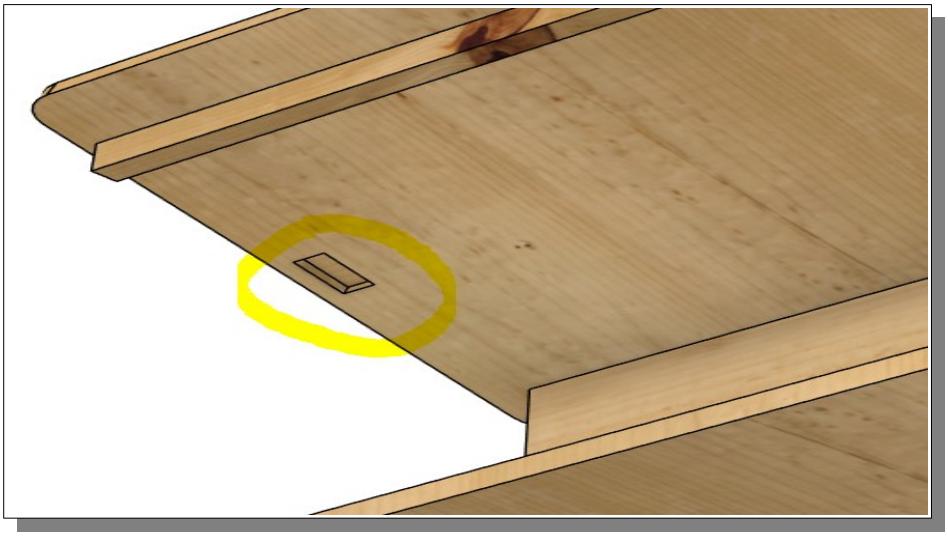


Again, start by screwing the tabs in the back (to the right hand side) and the base (again to the Right hand side). Follow this with the 2 screws in the base up into the RHS and the 2 screws in the face/rear edge of the RHS into the back.

Follow this with the door jamb. It has a small rebate on each end, this faces “outwards” (towards the doors). It is also rounded on the bottom edge. There are 2 screws, one in each end. Carefully align the jamb making sure the outside edge of the jamb is flush with the sides and sitting down in its notch. You should be able to tighten these screws flush with the face of the rebates.



The next piece is the top which has 2 locating slots that engage with the mini tabs on the top of the sides. As shown below



Fit the top and check it is sitting down on the sides and touching at the back. There are 13 screws in the top – this is done to assist in stopping water ingress. You can tighten them all at this point. The basic box build is now complete. Have a bit of a look to check you haven't left out a screw.

The last step before painting is to screw the handles onto the doors (unless you are planning to paint the handles a different colour). The handles are pre-drilled and use the same screws as the box. They are designed to be oriented vertically. If you want, you can put you own selection of handle on the box.

Painting

TG3D recommends that you paint the library at this point. Give it a light sand using 120 grit and then paint it using a quality exterior (outdoor) paint in colour(s) of your choice. Paint the inside as well as both sides of the doors. Be generous about painting around the joints in the box to help keep water out.

Doors

Once your paint is dry, you can assemble to doors to the box. The 2 doors are different. The left door has a rebate on the inside that is designed to overlap the rebate on the right hand side door. The first step is to put the windows into the doors. Remove the protective covers on the windows and place them in the recesses in the inside of the door. There are 6 4G x 12mm screws in each window.

The hinge mounting holes are marked with "pop" marks. The screws are in the packet with the hinge. The hinges mounting is designed so the "hinge pin" sits in the recess on the front edge of the side. Due to the geometry, and the need to avoid screwing into the ends of the plywood, the hinges attached to the sides, are screwed in from the back of the hinge whereas the hinge side on the doors attaches from the hinge front. Select a side and place the hinge on using the top and bottom screws. I will have drilled pilot holes to assist in getting the screws straight and in the right position. Using a #2 phillips driver, they should be tight but be careful, these are only small screws. You should then be able to offer up the door and again, put the top and bottom screws, on the inside of the door using the pop marks and pilot holes. Repeat this for the other side. The doors should close with about 1mm of clearance between them. Put the rest of the screws into both hinges.

Screw the 2 handles to the front of the door using the pre-drilled holes.

Catch

The last job is to fit the catch. Use the screws supplied. I supply 2 catches but the catch only needs to be on the left hand door. The right hand door is held close by the rebate on its inner edge against the left hand door. You can install the right hand catch if you need to.

There are “pop” marks on the inside of the left hand door and on the top of the base for the catch. Screw the catch, loosely on to the base and the striker onto the door. Close the door and engage the catch. Adjust the position of the catch so the door is closed. Tighten the screws in the catch and check that the left and right hand doors are held closed and the doors are easily opened.

At this point the box is assembled. Congratulations! Check all the sides for missing screws, there are 50 of them! I include a few extra screws so don’t be concerned about having extra.

Mounting

There aren’t many rules about mounting this box. Some suggestions include

- Screw it to your fence through the back.
- Get a “wing split” treated pine post, concrete it into a hole and again screw through the back to the post

The library is quite strong but if you are worried about kids hanging of it, a shelf support bracket underneath it would be sufficient.

If you enjoyed making and using this street library, please give me a like or follow at

https://facebook.com/tg3ddesign

I’d appreciate a review, if possible, as well as a few photos for my gallery?

Thanks in advance

Tony (TG3D)

Note: a full colour version of this manual is available at

https://tg3d.com.au/resources